ECE1512 Project A

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I. Introduction

For many real-life machine learning (ML) applications, the available training data comprises of multiple instances of an observation but with only one label for the entire bag of instances. Multiple instance learning (MIL) works with such weakly labeled data to (1) learn a model that predicts a bag label and (2) be able to identify instances that are most relevant to the prediction, providing some interpretability.

II. Literature Review

A. Attention-based Deep Multiple Instance Learning (ABMIL) [1]

1) Problem: Assuming that instances in each bag are independent and that the bag label is invariant to their ordering, the MIL problem can be decomposed into three general steps:

(i) feature extraction - embedding of instances into feature space
(ii) pooling/aggregation - permutation-invariant combination of instance embeddings, producing the bag embedding
(iii) prediction - linear projection of bag embedding to logits and conversion to probabilities

ABMIL addresses the limitations of predefined pooling in step (ii) for binary classification: predefined pooling operators (e.g. `max`, `mean`) cannot adapt to different types of data and provide no interpretability.

2) Contributions:

(a) An gated attention pooling mechanism that learns the relevance of each instance to the overall bag prediction, allowing the MIL model to adapt to different types of data and providing some interpretability through the attention scores (relevance). The two gate layers introduce more nonlinearity (as compared with just one layer) to the pooling mechanism, improving its expressiveness.
(b) Empirical results on classical MIL datasets (MUSK1, MUSK2, FOX, TIGER, ELEPHANT), MNIST-BAGS, and histopathology datasets (BREAST CANCER, COLON CANCER). ABMIL was shown to be comparable with the best performing classical MIL methods on classical datasets, outperform them on the small-sample size MNIST-BAGS dataset, and achieve the highest recall on the histopathology datasets. Some examples of strong correlation between attention scores and bag prediction were also presented.

B. Clustering-Constrained-Attention Multiple Instance Learning (CLAM) [2]

1) Problem: Attention pooling alone fails to exploit instance-level information.

2) Contributions:

(a) Multi-class aggregation architecture with an attention branch for each class, with each branch extending AB-MIL attention pooling with an additional clustering layer. For each class, the highest- and lowest-attended instances are labeled as positive and negative evidence respectively for that class. These instance-level pseudo-labels then provide a clustering loss against which a separating hyperplane in feature space can be learnt. This hyperplane thus predicts whether each instance gives evidence for that class, providing an auxiliary signal for feature discrimination. Additionally, if classes are mutually exclusive, positive evidence for one class can be assumed to be negative evidence for another class.

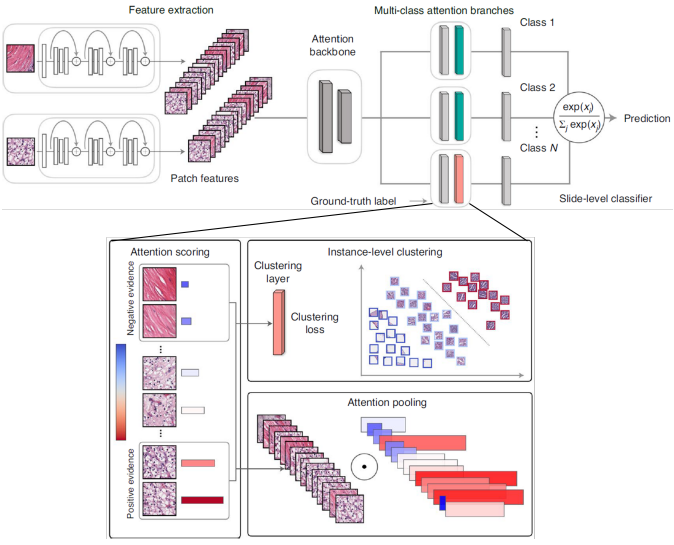


Fig. 1. CLAM multi-class aggregation architecture

(b) Rendering of attention scores as heatmaps overlayed on each slide.
(c) Empirical results on single and multi-class classification (renal cell carcinoma subtyping, non-small-cell lung cancer subtyping, lymph node metastasis detection) against the MIL max-pooling baseline. CLAM was shown to be robust to data scarcity, and generalise to independent

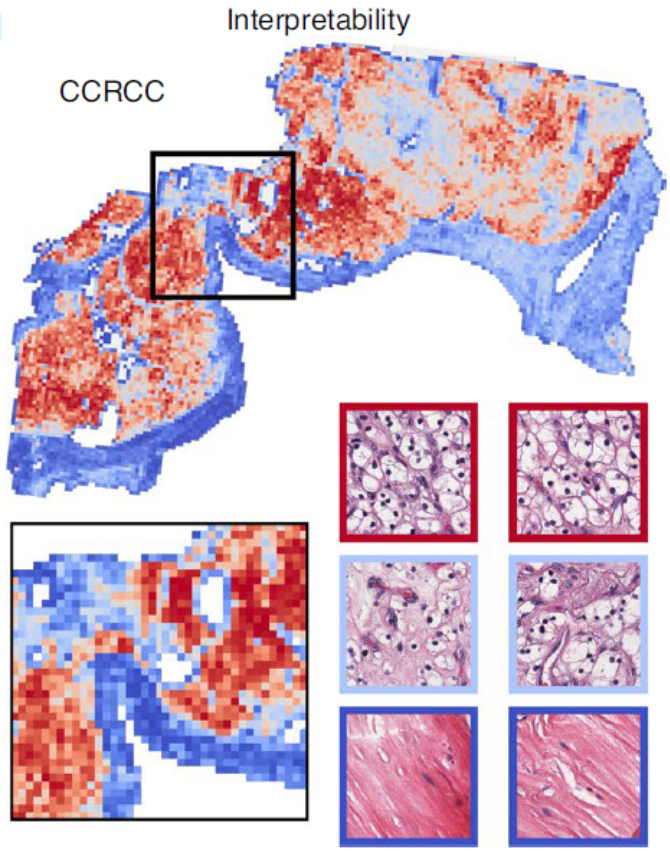


Fig. 2. CLAM heatmap

cohorts, including slides from other institutions, biopsy and resection specimens, and smartphone microscopy images.

(d) Ablation experiments across training sets of different sizes, observing that the additional instance-level supervision improves model accuracy over using bag-level supervision alone for small training sets.

(e) An open-source whole slide image (WSI) processing pipeline that segments each slide into patches (instances) of a standard size which are then fed through a pre-trained CNN feature extractor that maps them to a low-dimensional feature space, drastically reducing the dimensionality of the input to the attention layers.

C. Dual-Stream Multiple Instance Learning (DSMIL) [3]

1) Problem:

(a) MIL max-pooling aggregation strongly biases the decision boundary towards the most positive instances in each bag, presenting a high risk of overlooking less-positive instances.

(b) Poor instance features due to direct slide-level training of feature extractor on unbalanced positive bags which have a disproportionate number of negative instances.

2) Contributions:

(a) Dual-stream aggregator that jointly learns an instance classifier and a bag classifier. Stream 1 uses max-pooling on instance scores to find the critical instance and its score. Stream 2 computes attention weights from a distance measure between each instance's and the critical instance's query, then computes the bag embedding as the distance-weighted average of the instances' value. The bag score is then computed from a learnt linear map of the bag embedding. The final score is then the average of the scores from the two streams. The distance measure applies inter-instance selection, shifting the decision boundary according to how similar instances are to the critical instance. This proximity-weighting provides a regularisation, with respect to the critical instance, on the learned attentions. DSMIL aggregation can be extended to the multi-class case by having independent stream pairs for each class.

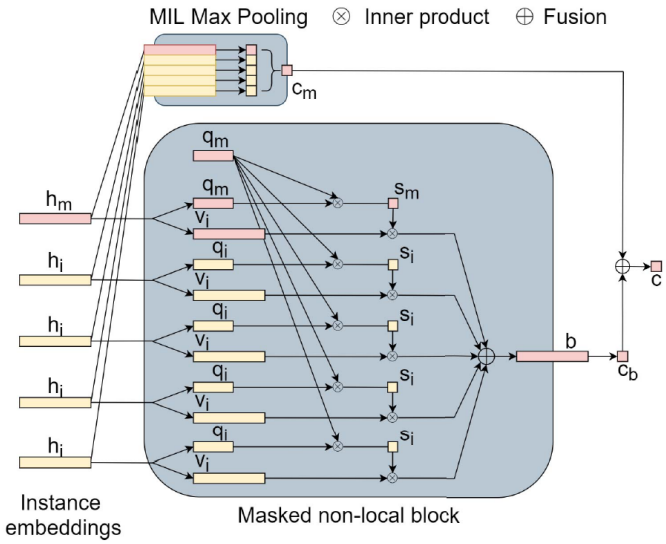


Fig. 3. dual-stream aggregator architecture

(b) Self-supervised contrastive learning (SimCLR) framework for feature extractor (CNN) training. SimCLR (Fig. 4) trains a different CNN at each scale to associate sub-images from the same image by randomly augmenting a random selection of sub-images and having the CNN maximise the agreement between sub-images from the same image.

(c) Pyramidal multiscale feature fusion (Fig. 4) that concatenates feature vectors of the same instance, each produced by a different CNN for each magnification. This preserves information from different scales and allows the aggregator to select the appropriate value based on query similarity across different scales.

(d) Empirical results on unbalanced/balanced and single/multi- class MIL datasets (Camelyon16 and TCGA lung cancer) against MIL baselines (mean-pooling, max-pooling, RNNMIL [4], ABMIL), and a upper-bound fully-supervised model with pixel-level

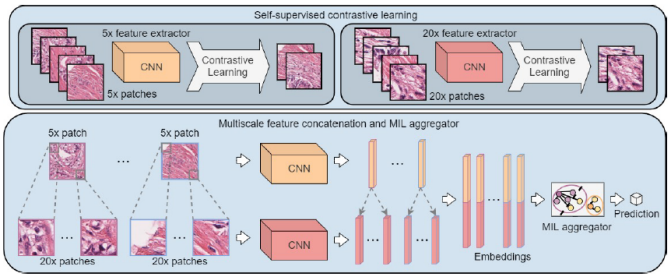


Fig. 4. Self-supervised Contrastive Learning (SimCLR) and Locally-Constrained Multiscale Attention (LC)

labels. DSMIL was shown to have at least 1% accuracy improvement over all baselines across both datasets, using both conventional patch-based features and SimCLR features, and both single and multi- scale features. For the unbalanced Camelyon16 dataset, DSMIL had about 3% AUC improvement over all baselines with both single and multi- scale features, and fell short of the fully-supervised model (single scale) by about 4% AUC.

(e) Empirical results on classical MIL datasets (MUSK1, MUSK2, FOX, TIGER, ELEPHANT) against non-local aggregators (NLMIL [5], ANLMIL[1] [6]). The dual-stream aggregator had an average 3% improvement over non-local aggregators on classical MIL problems.

(f) Ablation studies with fixed DSMIL aggregator show at least 16% higher classification accuracy with SimCLR on unbalanced bags (Camelyon16) as compared to end-to-end max-pooling trained feature extractors. However, for balanced bags, patch-based methods outperformed SimCLR.

Ablation on multiscale attention (DSMIL-LC) on unbalanced data showed an accuracy improvement of 3% over single scale approaches and at least 1.5% improvement over multiscale approaches (concatenation, max-pooling, mix). However, using two scales ($5_\times$,$20_\times$) produced better results than three scales ($1.25_\times$,$5_\times$,$20_\times$) as coarse scale similarities might cause the inter-instance selection to be less discriminate.

D. Transformer based Correlated Multiple Instance Learning for Whole Slide Image Classification (TransMIL) [7]

1) Problem: MIL methods typically assume instances within a bag are independently and identically distributed (i.i.d.) and neglect correlations between instances. However, for WSI pathology, pathologists often consider both the contextual information around a single area and the correlations between different areas when making a diagnostic decision.

2) Contributions:

(a) Generic multi-class correlated MIL architecture with convergence proof. This architecture adapts the ABMIL formulation to include correlation between instances. Instead of only capturing independent morphological features of each instance, TransMIL also captures morphological correlations and spatial information across the entire bag of instances.

(b) Transformer-based MIL (TransMIL) implementation of (a). TransMIL (Fig. 5) first employs the pretrained ResNet50 network to extract morphological features from (non-background) bag instances in an ordered sequence, potentially preserving some spatial context. The sequence of feature embeddings is then padded with embeddings from the front of the sequence until the total number of embeddings is a square number. This squaring of the sequence allows for the sequence to be later reshaped[2] into a 2D grid for positional encoding. The sequence of squared embeddings is then concatenated with the class embedding[3] and fed through a Transformer-PPEG-Transformer (TPT) module where the multi-head self-attention (MSA) transformers aggregate morphological information, and the Pyramid Position Encoding Generator (PPEG) encodes spatial information of only the squared embeddings (Fig. 6). The self-attention is approximated with the Nyström method [8] to reduce the memory and time complexity from $O(n^2)$ to $O(n)$. The PPEG fuses different sized kernel convolutions, producing an encoding that potentially captures spatial information at various scales. The class embedding after the final MSA transformation is then normalised and fed to the Multilayer Perceptron (MLP) head to make the final prediction.

Note that since tissue samples are irregular, background patches are not captured in the embedding sequence, and the varying-length sequences are padded to square, the resulting square arrangement of instances is unlikely to represent the true spatial relation between instances.

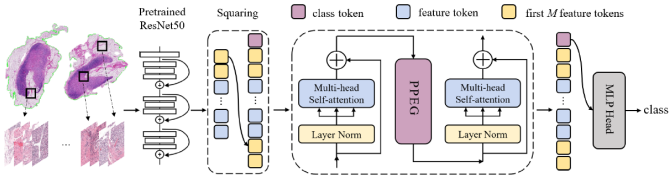


Fig. 5. TransMIL Architecture

(c) Heatmap visualisation of TransMIL attention scores with limited examples of strong correlation between attention scores and annotated ground truth.

(d) Empirical results on CAMELYON16, TCGA-NSCLC, and TCGA-RCC datasets at $20_\times$ magnification, against mean-pooling, max-pooling, ABMIL, PT-MTA, RNN-MIL, DSMIL, CLAM-SB, CLAM-MB. TransMIL shown to outperform ABMIL, PT-MTA, and CLAM, which neglect correlation between instances, by at least 5% AUC on CAMELYON16. TransMIL was also more accurate

[1]DSMIL Stream 1 instance scores, and Stream 2 query and value scores are computed using different mappings.

[2]squaring is a canonical choice for a rectangular grid independent of the number of tokens in the sequence.

[3]learnt embedding for the class, identical for all slides.

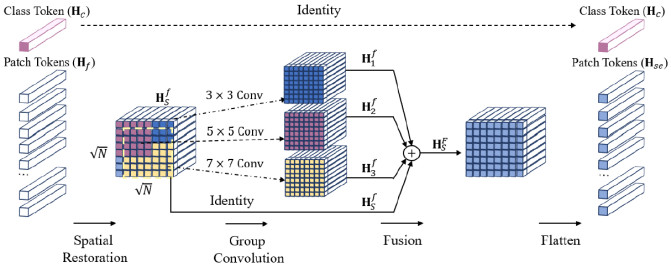


Fig. 6. TransMIL PPEG

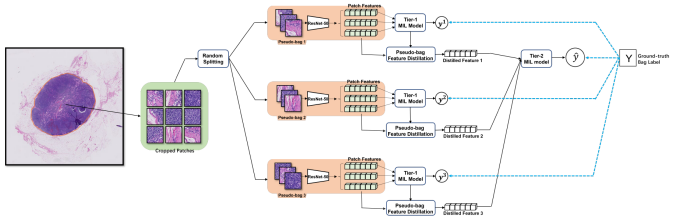


Fig. 7. DTFD-MIL Architecture

and converged faster than all baselines over all the unbalanced/balanced and binary/multiple classification tasks.

(e) Ablation experiments showing that PPEG provides marginal improvement over sinusoidal (absolute) position encoding.

E. Double-Tier Feature Distillation Multiple Instance Learning for Histopathology Whole Slide Image Classification (DTFD-MIL) [9]

1) Problem:

(a) Rigorous link between instance attention scores under ABMIL frameworks and their classification probabilities had yet to be established.

(b) Overfitting in WSI pathology MIL due to inherent dataset characteristics — limited slides and small proportion of positive instances — not addressed by other MIL approaches.

2) Contributions:

(a) Instance probability derivation for ABMIL shown to produce probabilities that are more reliable than raw attention scores in some cases.

(b) Double-tier feature distillation framework (DTFD-MIL) with feature distillation derived from instance probabilities (Fig. 7):

1. split each bag (slide) into smaller pseudo-bags, each inheriting their parent's label. This virtually multiplies the number of training bags and reduces the imbalance of instances in each bag. Introduces label noise from the creation of false positive pseudo-bags
2. Tier 1: ABMIL model on each pseudo-bag
3. pseudo-bag feature distillation from instance(s) selected based on one of four strategies:
   - `MaxS`: highest probability instance
   - `MaxMinS`: highest- and lowest-probability instances, concatenated
   - `MAS`: highest-attention instance
   - `AFS`: attention-aggregated feature of the whole pseudo-bag
4. Tier 2: second ABMIL model aggregates the distilled features into a prediction for the parent slide

(c) Empirical results on CAMELYON16, TCGA (Lung Cancer) against mean-pooling, max-pooling, RNN-MIL, ABMIL, DSMIL, CLAM-SB, CLAM-MB, TransMIL. On CAMELYON16, DTFD-MIL across all distillation strategies outperformed all baselines, with `AFS` being the best distillation strategy with at least a 4% AUC over all baselines. DTFD-MIL regardless of distillation strategy had a model size and FLOPs comparable to most baselines, and significantly smaller than TransMIL. Improvement over baselines for TCGA was smaller due to instance-level methods performing well on slides with significantly larger positive regions. Nonetheless, the best performing model on TCGA was DTFD-MIL with `MaxMinS` with AUC of 96.1

(d) Ablation experiments showing that increasing the number of pseudo-bags improves AUC for both tiers up to some point. Tier-2 generally outperforms Tier-1, especially on CAMELYON-16, which supports the motivation for the second tier on unbalanced bags. `MaxS` is the weakest strategy for both datasets.

III. Research Questions

1) **feature extractor fine tuning**: can instance-level supervision (II-B) be used to train a foundation feature extractor (for a specific task)? How does this compare with end-to-end training with regards to accuracy and computational cost?
2) **self-supervised contrastive learning**: for bags with a disproportionate number of positive instances, would SimCLR (II-C) introduce a positive bias on negative instances and how does that affect accuracy?
3) **attention along magnification**: would attention along magnification help improve multiscale feature fusion with coarser scales?
4) **TransMIL**
   - how is the TransMIL class embedding learnt and how would other MIL baselines perform if they used the same class embedding for regularisation?
   - how can TransMIL be adapted to handle higher magnifications when the model size and compute scales with the number of instances per bag?
5) **DTFD-MIL**
   - how should the number of pseudo-bags be varied given the degree of imbalance in each bag? What is the relationship between label noise and performance?
   - how does the derived instance probability compare with raw attention scores in terms of accuracy across all instances?

- why does `AFS`, which does not use instance probabilities, perform best on CAMELYON-16 while `MaxS` performs worst?

The code for the following sections can be found at https://github.com/strangehiccups/ECE1512-Project_A

IV. Baseline Summary

A. Model Structure

The baseline ABMIL model takes feature embeddings as input and consists of 3 components:

1) **dimension reduction**: single linear layer + ReLU activation encoder maps input feature embeddings to lower dimensional space
2) **gated attention pooling**: two linear gate layers, one with tanh and the other with sigmoid activation + linear map from the product of the gate outputs to each class (attention logits). The attention logits are then normalised to their probability simplex and used to weight the output embeddings from 1). Aggregating over all instances then produces the bag embedding.
3) **prediction**: dropout + linear map classifier projects bag embedding to class logits

The final prediction is then simply the class with the largest logit.

B. Dataset and DataLoader

- A `Dataset` defines the structure of the data and where to read it from while a `DataLoader` defines how the data is fed to a model.
- The `DataLoader` for training should use `shuffle=True` as all modern ML models learn through stochastic gradient descent, where each batch gradient is meant to be a noisy but unbiased estimate of the gradient over the whole dataset. Shuffling batches (in this case slides) every epoch is required to avoid cycling trajectories which mitigates order memorisation and improves exploration.
  The `DataLoader` for validation and test do not need to shuffle batches as the model is not learning during evaluation but simply producing a prediction for each sample. However, `shuffle=` is often set to `False` for evaluation to keep the sample order consistent across different settings so that multiple evaluations can be easily compared.
- For a batch in this project with input shape `[B,N,D]`
  - `B`: number of slides[4] (bags) in the batch
  - `N`: number of patches (instances) in each slide
  - `D`: feature embedding dimension

[4] `B=1` for all batches for all datasets (BRACS, CAMELYON16, CAMELYON17) in this project. This is likely a convenient standard for WSI datasets where slides have different number of patches.

C. Training Procedure

The train loader cycles through all batches randomly each epoch with the model and optimiser processing one batch at a time, for a total of 30 epochs. During each epoch:

1) optimiser resets gradients to zero (see IV-E)
2) model makes prediction on the batch
3) loss function (cross entropy) computes the loss. Cross-entropy loss measures the difference between the distribution predicted by the model and the true label distribution
4) gradients are backpropagated
5) optimiser updates the model parameters according to the gradients to minimise the loss

D. ABMIL Implementation

- All linear module (`nn.Linear`) weights are trainable parameters.
- All attention weights after softmax are non-negative and their sum is 1.
- We can use any (at least weakly order-preserving) function $f : \mathbb{R}^n \to \Delta^{n-1}$ that maps the attention weights $\mathbf{z} \in \mathbb{R}^n$ to their probability simplex, e.g. Sparsemax [10]:

$$Sparsemax(\mathbf{z}) := \underset{\mathbf{p}\in\Delta^{n-1}}{\arg\min} \|\mathbf{p}-\mathbf{z}\|^2, \quad \mathbf{z} \in \mathbb{R}^n$$

  Sparsemax in this case allows the aggregation to eliminate instances that are completely irrelevant to the classification.

E. Validation and Test Evaluation

- `model.eval()` turns off training related behaviour in all layers of the model e.g.:
  - dropout which is meant to be a regularisation technique to mitigate overfitting. Evaluating a model with dropout active is equivalent to evaluating a randomly different (smaller) model.
  - batch norm tracking of current statistics as there is no longer a need for gradients to flow through them during evaluation

  and does not affect gradient tracking.
- `torch.no_grad()` turns off gradient tracking which saves on memory and compute during evaluation when the model is not learning. Gradient tracking is performed by the gradient computation engine autograd.
- If `optimizer.zero_grad()` was omitted, gradients would accumulate across batches (and epochs) and the effective gradient would no longer represent the actual gradient for each batch. Since the gradient accumulates, the effective learning rate also grows with the number of iterations which would eventually lead to unstable learning and blowing up of the loss.
- The test set is reserved for evaluating the model design, including its ability to generalise to unseen data.

V. Baseline Evaluation

A. Configuration

The baseline ABMIL model was trained, validated, and evaluated on precomputed feature embeddings from three datasets (CAMELYON16, CAMELYON17, BRACS), on an RTX 5070, and with the following hyperparameters:

- inner dimension: `256`
- attention dimension: `128`
- dropout:
  - CAMELYON16: `0`
  - CAMELYON17, BRACS: `0.3`
- loss function: `CrossEntropyLoss`
- optimiser: `AdamW`
  - learning rate: `1e-4`
  - weight decay: `1e-1`
- training epochs: `30`

With `AdamW` learning rate at `1e-4`, the weight decay was set to `1e-1` to produce an overall weight reduction of about 7-11% over 30 epochs (with about 240 to 400 batches per epoch across the three datasets) which is significant but not too extreme.
Dropout was neglected for CAMELYON16 as it is a simple binary classification task and there is a low risk of overfitting. To reduce seed variance while keeping training times feasible, 5 runs[5] were performed for each dataset. The mean and standard deviation of the learning curves and all test metrics are reported in V-B and VII-B.

B. Results

The ABMIL model performed well on the CAMELYON16 dataset, converging in about 11 epochs without overfitting as shown in Fig. 8, and with high **AUC** and $\mathbf{F_1}$ scores as shown in Table I.
However, it performed poorer on CAMELYON17, overfitting at about epoch 10 (Fig. 9), and poorer still on BRACS, overfitting almost immediately (Fig. 10) even with dropout. Coupled with training losses converging close to 0, the learning curves suggest that there was a significant degree of memorisation on the training set. Memorisation is also consistent with the lower **AUC** and significantly lower $\mathbf{F_1}$ scores for these two datasets.

TABLE I
Baseline Results[8] on CAMELYON16

Model	AUC	Accuracy (%)	$\mathbf{F_1}$	Loss
ABMIL	$0.970_{0.008}$	$95.7_{0.425}$	$0.953_{0.005}$	$0.303_{0.035}$

[5]seeds: 42, 15, 12, 17, 73
[7]normal slide, no annotation
[8]subscripts are the corresponding standard deviations.

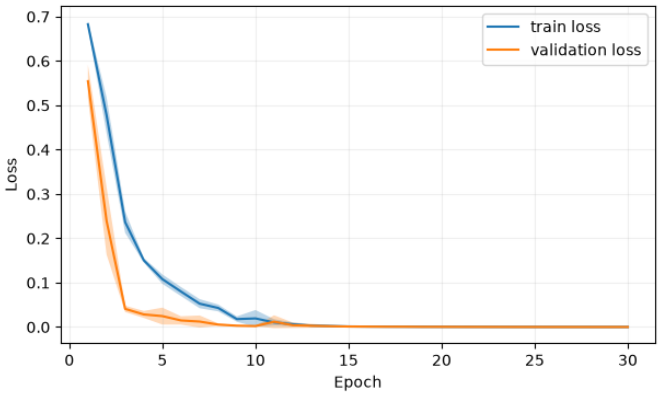


Fig. 8. ABMIL learning curves on CAMELYON16

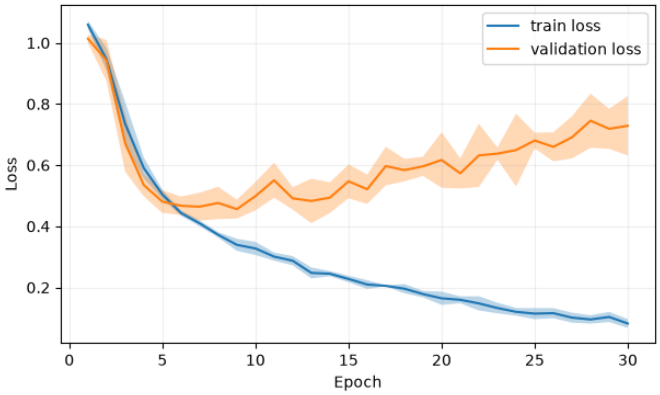


Fig. 9. ABMIL learning curves on CAMELYON17

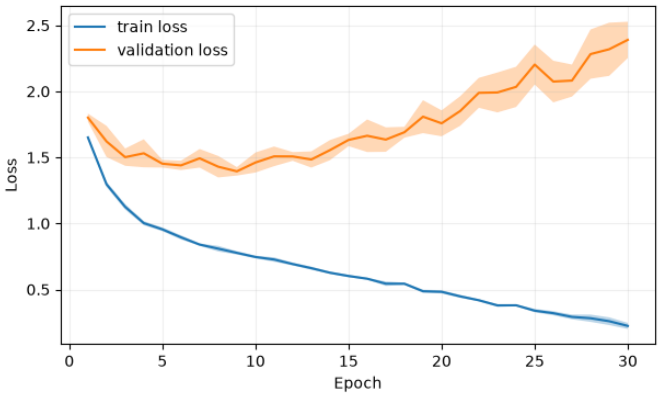


Fig. 10. ABMIL learning curves on BRACS

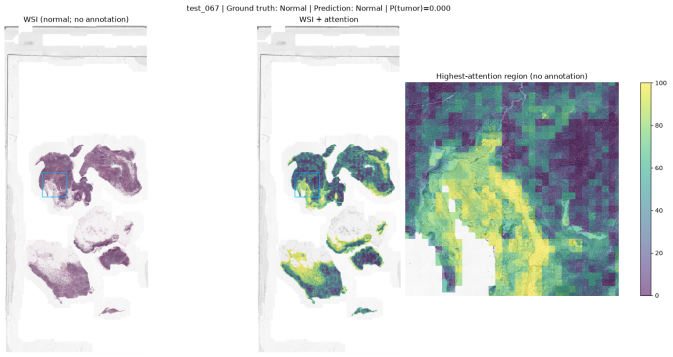


Fig. 11. ABMIL (`seed=42`) attention heatmap[7] on CAMELYON16 (test_067)

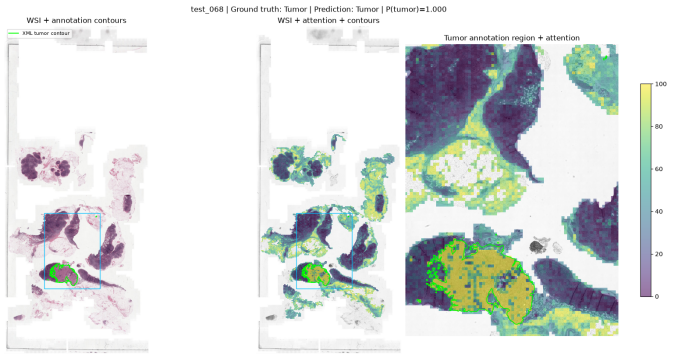


Fig. 12. ABMIL (`seed=42`) attention heatmap vs ground truth annotations on CAMELYON16 (test_068)

VI. Research Ideas

A. Multi-Class Attention

Since the baseline model's single attention map must serve multiple classes for CAMELYON17 and BRACS, it is possible that, instead of learning class-specific features, the model was forced to learn slide-specific features that happen to separate slides along the classes, leading to the observed memorisation. Thus, introducing a separate attention map for each class (as in II-B) may improve the model's ability to generalise with multi-class datasets.

The additional parameters from the attention maps scale with the number of classes which is much smaller than the inner and attention dimensions. Thus, introducing multi-class attention only adds about 2% and 7% (for CAMELYON17 and BRACS respectively) more parameters to the baseline model.

B. Graph Representation with Knowledge-Aware Attention

The baseline ABMIL model also does not make use of inter-instance correlations which are likely relevant for CAMELYON17 where the severity is defined by the size of the largest focus, and for BRACS where subtypes depend on how cells are arranged and on the pattern and extent of the lesion, which span multiple patches. As discussed in II-D, inter-instance correlation could be incorporated by preserving the spatial relations between patches and extracting this information using a CNN. However, due to tissue samples being irregular and rejection of background patches, a simple sequence does not reliably preserve spatial relations. It is also impractical to additionally store spatial relations between large numbers of patches.

Instead, inter-instance relations could be deduced from patch embeddings. One such approach (WiKG) proposed by Li et al. [11] constructs a knowledge graph representation of each slide, in which every patch is a node and each link between two patches is a directed edge. Each patch's neighbours are determined as its k most similar patches. Knowledge-aware attention then weighs how much each neighbour should contribute to a patch embedding, taking into account the patch, the neighbour, and the edge between them. The weighted neighbour information is fused with the patch's own embedding, so correlated patches share information before the slide-level prediction.

Since the top-k patches are computed for every patch, the size of the graph representation scales with the number of patches, introducing a significant memory and computation cost for WSIs.

VII. Research Extension

Since the main problem with the baseline is overfitting on CAMELYON17 and BRACS, it would be difficult to investigate the impact of the WiKG approach without first addressing the lack of separate attention maps for each class. The slide-level attention aggregation in the baseline model was expanded to have an attention map per class for CAMELYON17 and BRACS.

A. Configuration

The resulting multi-class ABMIL (MC-ABMIL) model was trained, validated, and evaluated with the same baseline hyperparameters and on the same hardware to isolate the difference in the models to the addition of separate attention maps.

B. Results

For CAMELYON17, MC-ABMIL outperformed ABMIL with ∼0.04 **AUC** and ∼0.15 $\mathbf{F_1}$ improvement (Table II), with slightly less overfitting (Fig. 13).

However, on BRACS, there wasn't a significant improvement to the performance metrics (Table III), and overfitting appears to be more severe (Fig. 14). This is possibly due to the following characteristics that distinguish BRACS from the other two datasets:

1) small training set of 389 slides for 7 classes. Rare subtypes may only be present in a handful of examples which may not be enough data to learn from. This can be addressed at the source (obtain more data) or with data augmentation.
2) the slide label is the most severe lesion while the task is subtyping. The ABMIL model lacks an explicit class ranking mechanism and thus may not be able to distinguish between the strong presence of a subtype and

its impact on the label[9]. Adding more aggregators only improves detection, not classification.

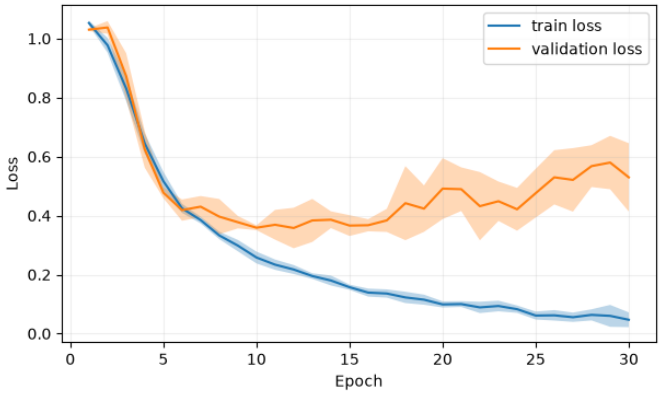


Fig. 13. MC-ABMIL learning curves on CAMELYON17

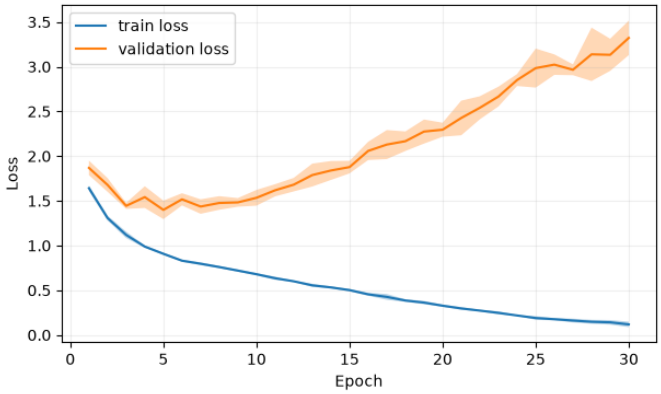


Fig. 14. MC-ABMIL learning curves on BRACS

TABLE II
RESULTS ON CAMELYON17

Model	AUC	Accuracy (%)	F_1	Loss
ABMIL	$0.857_{0.023}$	$80.1_{2.655}$	$0.533_{0.023}$	$0.611_{0.107}$
MC-ABMIL	$0.898_{0.003}$	$82.4_{2.302}$	$0.676_{0.013}$	$0.544_{0.029}$

TABLE III
RESULTS ON BRACS

Model	AUC	Accuracy (%)	F_1	Loss
ABMIL	$0.802_{0.012}$	$44.7_{3.142}$	$0.353_{0.036}$	$1.512_{0.275}$
MC-ABMIL	$0.798_{0.006}$	$42.8_{3.227}$	$0.354_{0.045}$	$1.437_{0.042}$

[9]For CAMELYON17, the strong presence of metastases correlates with the severity label

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